

VIVEK LINUX COMMAND CHEATSHEET

1. Linux Common Commands

Commands	Short Description / use
pwd	# Show current working directory
ls	# List files and folders
ls -l	# List with permissions and details
ls -la	# List all files including hidden files
ls -lh	# Show file sizes in human-readable format
cd /var/log	# Change directory
cd ..	# Go one directory back
cd ~	# Go to home directory
cd -	# Go to previous directory
clear	# Clear terminal
history	# Show command history
history tail	# Show last commands
whoami	# Show current user
hostname	# Show system hostname
date	# Show current date and time
uptime	# Show system uptime

2. Files & Directories

Commands	Short Description / use
mkdir project	# Create directory
mkdir -p project/app/logs	# Create nested directories
touch app.log	# Create empty file
cp app.log backup.log	# Copy file
cp -r app/ backup/	# Copy directory
mv old.txt new.txt	# Rename file
mv file.txt /tmp/	# Move file
rm file.txt	# Delete file
rm -r project/	# Delete directory
rm -rf project/	# ⚠ Force delete directory

file app.log	# Identify file type
stat app.log	# Show detailed file information
tree	# Display directory structure
tree -L 2	# Display 2 directory levels

3. Read & Search Files

Commands	Short Description / use
cat app.log	# Display entire file
less app.log	# Read file page by page
head app.log	# Show first 10 lines
head -n 20 app.log	# Show first 20 lines
tail app.log	# Show last 10 lines
tail -n 50 app.log	# Show last 50 lines
tail -f app.log	# Follow live log updates
grep "ERROR" app.log	# Search for ERROR
grep -i "error" app.log	# Case-insensitive search
grep -r "ERROR" /var/log/	# Search recursively
grep -n "ERROR" app.log	# Show matching line numbers
find /var/log -name "*.log"	# Find log files
find . -type f	# Find files
find . -type d	# Find directories
which python3	# Find executable location
whereis python3	# Find binary/manual locations

4. Permissions

Commands	Short Description / use
ls -l	# View permissions
chmod +x script.sh	# Make script executable
chmod 755 script.sh	# Owner=rwx, others=rx
chmod 644 file.txt	# Owner=rw, others=r
chmod 600 secret.txt	# Owner only read/write
chmod -R 755 project/	# Apply permission recursively
chown ubuntu file.txt	# Change owner
chown ubuntu:ubuntu file.txt	# Change owner and group

chown -R ubuntu:ubuntu project/ # Change recursively

umask # Show default permission mask

5. Users & Groups

id # Show user/group IDs

groups # Show current user's groups

who # Show logged-in users

w # Show users and activity

sudo -i # Switch to root shell

su - username # Switch user

sudo useradd -m devops # Create user with home directory

sudo passwd devops # Set user password

sudo usermod -aG docker devops # Add user to docker group

sudo groupadd developers # Create group

sudo userdel -r devops # ⚠ Delete user and home directory

6. Processes

Commands	Short Description / use
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ps aux	# Show all running processes
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ps -ef	# Detailed process list
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top	# Live process monitoring
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htop	# Interactive process monitor
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pgrep nginx	# Find nginx process ID
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pidof nginx	# Show nginx PID
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kill 1234	# Gracefully stop process
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kill -9 1234	# ⚠ Force kill process
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pkill nginx	# Kill processes by name
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jobs	# Show background jobs
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bg	# Resume job in background
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fg	# Bring job to foreground
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nohup ./app.sh &	# Run process after logout
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7. Disk & Memory

Commands	Short Description / use
df -h	# Show disk usage
du -sh /var/log	# Show directory size
du -sh *	# Show size of items in current directory
free -h	# Show RAM usage
lsblk	# Show disks and partitions
blkid	# Show block device UUID/type
mount	# Show mounted filesystems
uname -a	# Show kernel/system information
cat /etc/os-release	# Show Linux distribution
lscpu	# Show CPU information
uptime	# Show load and uptime

8. Networking — VERY IMPORTANT FOR DEVOPS

Commands	Short Description / use
ip addr	# Show IP addresses
ip link	# Show network interfaces
ip route	# Show routing table
hostname -I	# Show system IP
ping -c 4 google.com	# Test connectivity
curl https://example.com	# Make HTTP request
curl -I https://example.com	# Show HTTP headers
wget https://example.com/file.zip	# Download file
ss -tulnp	# Show listening ports
ss -ant	# Show TCP connections
nc -zv localhost 8080	# Test whether port is open
traceroute google.com	# Trace network path
dig google.com	# DNS lookup
nslookup google.com	# DNS lookup
host google.com	# DNS information

9. SSH & EC2

Commands	Short Description / use
ssh user@192.168.1.10	# Connect to remote Linux server
chmod 400 key.pem	# Secure private key
ssh -i key.pem ubuntu@EC2_PUBLIC_IP	# Connect to Ubuntu EC2
ssh -i key.pem ec2-user@EC2_PUBLIC_IP	# Connect to Amazon Linux EC2
scp file.txt user@server:/tmp/	# Copy file to server
scp user@server:/tmp/file.txt .	# Download file from server
scp -r project/ user@server:/home/user/	# Copy directory
rsync -av project/ user@server:/app/	# Efficient directory synchronization

10. Package Management

Commands	Short Description / use
sudo apt update	# Update package index
sudo apt upgrade	# Upgrade installed packages
sudo apt install nginx	# Install package
sudo apt remove nginx	# Remove package
sudo apt purge nginx	# Remove package + configuration
sudo apt autoremove	# Remove unused packages
apt search nginx	# Search packages
dpkg -l	# List installed packages

Amazon Linux/RHEL

Commands	Short Description / use
sudo dnf update	# Update packages
sudo dnf install nginx	# Install package
sudo dnf remove nginx	# Remove package
sudo dnf search nginx	# Search package
rpm -qa	# List installed RPM packages

11. Services / systemctl

Commands	Short Description / use
systemctl status nginx	# Check service status
sudo systemctl start nginx	# Start service
sudo systemctl stop nginx	# Stop service
sudo systemctl restart nginx	# Restart service
sudo systemctl reload nginx	# Reload configuration
sudo systemctl enable nginx	# Start automatically at boot
sudo systemctl disable nginx	# Disable automatic startup
systemctl is-active nginx	# Check if service is active
systemctl is-enabled nginx	# Check boot configuration
systemctl --failed	# Show failed services

12. Logs — Very Important for DevOps

Commands	Short Description / use
journalctl	# View system logs
journalctl -u nginx	# View nginx logs
journalctl -u nginx -f	# Follow nginx logs
journalctl -f	# Follow system logs
journalctl -xe	# Detailed recent errors
journalctl -p err -b	# Show boot errors
dmesg	# Kernel messages
tail -f /var/log/syslog	# Follow Ubuntu system log
tail -f /var/log/auth.log	# Follow authentication logs

13. Pipes & Redirection

Commands	Short Description / use
ls -l grep ".log"	# Send ls output to grep
cat app.log grep ERROR	# Search errors using pipe
command > output.txt	# Save output to file
command >> output.txt	# Append output
command 2> error.txt	# Save errors
command > output.txt 2>&1	# Save output + errors
command &> output.txt	# Save both output and errors

command1 && command2	# Run command2 if command1 succeeds
command1 command2	# Run command2 if command1 fails

14. Text Processing

Commands	Short Description / use
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wc -l app.log	# Count lines
wc -w app.log	# Count words
sort names.txt	# Sort lines
uniq names.txt	# Remove consecutive duplicates
sort names.txt uniq	# Sort and remove duplicates
cut -d: -f1 /etc/passwd	# Extract first field
awk '{print \$1}' access.log	# Print first column
sed 's/old/new/g' file.txt	# Replace text
xargs	# Build commands from input

15. Compression

Commands	Short Description / use
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tar -cvf backup.tar project/	# Create TAR archive
tar -xvf backup.tar	# Extract TAR archive
tar -czvf backup.tar.gz project/	# Create compressed TAR
tar -xzvf backup.tar.gz	# Extract compressed TAR
gzip file.txt	# Compress file
gunzip file.txt.gz	# Decompress file
zip -r project.zip project/	# Create ZIP
unzip project.zip	# Extract ZIP

16. Cron Jobs

Commands	Short Description / use
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crontab -e	# Edit cron jobs
crontab -l	# List cron jobs
crontab -r	# ⚠ Remove cron jobs

DevOps Tools

git --version	# Check Git version
git init	# Initialize repository
git clone URL	# Clone repository
git status	# Show repository status
git add .	# Stage all changes
git commit -m "message"	# Create commit
git log --oneline	# Show compact commit history
git branch	# List branches
git switch -c feature	# Create and switch branch
git switch main	# Switch branch
git merge feature	# Merge branch
git pull	# Download and merge changes
git push	# Upload commits
git fetch --all	# Download remote information
git diff	# Show changes
git stash	# Temporarily save changes
git stash pop	# Restore stashed changes
git remote -v	# Show remote repositories

Docker

docker --version	# Check Docker version
docker ps	# Show running containers
docker ps -a	# Show all containers
docker images	# List Docker images
docker pull nginx	# Download image
docker run nginx	# Run container
docker run -d --name web nginx	# Run container in background
docker run -d -p 8080:80 nginx	# Map host port to container port
docker stop web	# Stop container
docker start web	# Start container
docker restart web	# Restart container

<code>docker rm web</code>	<code># Remove container</code>
<code>docker rmi nginx</code>	<code># Remove image</code>
<code>docker logs web</code>	<code># View container logs</code>
<code>docker logs -f web</code>	<code># Follow logs</code>
<code>docker exec -it web bash</code>	<code># Enter container shell</code>
<code>docker inspect web</code>	<code># Show container details</code>
<code>docker stats</code>	<code># Monitor containers</code>
<code>docker build -t myapp:1.0 .</code>	<code># Build image</code>
<code>docker tag myapp:1.0 user/myapp:1.0</code>	<code># Tag image</code>
<code>docker push user/myapp:1.0</code>	<code># Push image</code>

AWS CLI

<code>aws --version</code>	<code># Check AWS CLI version</code>
<code>aws configure</code>	<code># Configure AWS CLI</code>
<code>aws configure list</code>	<code># Show AWS CLI configuration</code>
<code>aws sts get-caller-identity</code>	<code># Show current AWS identity</code>

EC2

<code>aws ec2 describe-instances</code>	<code># List EC2 instances</code>
<code>aws ec2 start-instances --instance-ids INSTANCE_ID</code>	<code># Start EC2 instance</code>
<code>aws ec2 stop-instances --instance-ids INSTANCE_ID</code>	<code># Stop EC2 instance</code>
<code>aws ec2 reboot-instances --instance-ids INSTANCE_ID</code>	<code># Reboot EC2 instance</code>

S3

<code>aws s3 ls</code>	<code># List S3 buckets</code>
<code>aws s3 ls s3://BUCKET_NAME</code>	<code># List bucket contents</code>
<code>aws s3 mb s3://BUCKET_NAME</code>	<code># Create bucket</code>
<code>aws s3 cp file.txt s3://BUCKET_NAME/</code>	<code># Upload file</code>

```
aws s3 cp s3://BUCKET_NAME/file.txt .    # Download file
aws s3 sync ./folder s3://BUCKET_NAME/    # Sync folder to S3
aws s3 rm s3://BUCKET_NAME/file.txt      # Delete S3 object
```

VPC

```
aws ec2 describe-vpcs                # List VPCs
aws ec2 describe-subnets            # List subnets
aws ec2 describe-route-tables        # List route tables
aws ec2 describe-internet-gateways   # List Internet Gateways
aws ec2 describe-nat-gateways        # List NAT Gateways
aws ec2 describe-security-groups     # List security groups
```

IAM

```
aws iam list-users                    # List IAM users
aws iam list-roles                    # List IAM roles
aws iam list-policies                 # List IAM policies
```